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Subject :-Introduction to Cyber Security

Unit -2 Cyber Crimes and Cyber Law

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- Defining Cybercrime
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Cyber Crime

Cybercrime refers to criminal activities carried out using computers and the internet, including hacking, data theft, malware attacks, and financial fraud.

Cybercrime refers to **illegal activities involving computers, networks, or the internet** as a tool to commit offenses. These crimes include:

- **Identity Theft** – Stealing personal information to commit fraud.
- **Financial Fraud** – Online scams, fake transactions, and credit card fraud.
- **Cyberbullying** – Harassment or threats through digital platforms.
- **Phishing Attacks** – Deceptive emails or websites tricking users into revealing sensitive data.
- **Hacking** – Unauthorized access to systems and data breaches.
- **Malware Attacks** – Spreading viruses, ransomware, and trojans to damage or steal data.

Classification of Cybercrime

1.Against Individuals: Crimes where individual persons are the victims.
These include e-mail spoofing, spamming, cyber defamation, cyber harassments and cyber stalking.

2. Against Property: These include credit card frauds, internet time theft and intellectual property crimes.

3. Against Organizations: These include unauthorized accessing of computer, denial Of service, computer contamination / virus attack, e-mail bombing, salami attack, logic bomb, trojan horse and data diddling.

4. Against Society: These include Forgery, Cyber Terrorism, Web Jacking.

Crime against Individuals

Cybercrime against individuals refers to **illegal activities committed using digital devices or the internet that target a specific person and cause personal, financial, psychological, or reputational harm**.

Against Individuals

1. **CyberStalking :- Cyberstalking** is the **repeated online harassment** or monitoring of an individual.

   Activities Involved- Threatening emails, Continuous messages , Tracking social media activity.

   Example Sending threatening messages repeatedly on WhatsApp or Instagram.

1. Phishing:- **Phishing** is a form of **social engineering attack** where attackers trick individuals into revealing sensitive information.

   Common Methods :- Fake bank emails, SMS scams (smishing) ,Voice calls (vishing)

   Example SMS saying *"Your bank account will be blocked, click here to verify"*.

1. **Email Spoofing (Individual Target):-** Forging the sender's email address to impersonate another person.

   Example Sending emails pretending to be a friend asking for money.

Against Individuals(Cont.)

4. **Defamation :-** Spreading **false or damaging information** about an individual online.

   Example:- Posting false allegations on social media to harm reputation.

5. **Cyberbullying** :- It involves **online harassment, humiliation, or abuse** using digital platforms.

   Forms Insulting comments, Spreading rumors, Sharing embarrassing content.

Against Property

1. **Cyber squatting** :- It is the illegal practice of registering, using, or selling a domain name that is identical or confusingly similar to a well-known brand, trademark, or personal name, with the intention of making profit or misleading users.

2. **Cyber vandalism:-** It is the intentional damage, defacement, deletion, or disruption of digital property such as websites, data, or computer systems without authorization, usually without direct financial gain.

3. **Intellectual Property Theft:-** Stealing or illegally copying copyrighted digital content.

4. **Transmitting virus:-** It refers to the intentional act of spreading malicious software (computer virus) from one system to another using digital means, without the consent or knowledge of the victim.

Against Government

1. **Cyber Terrorism:-** Using cyber attacks to threaten national security or create panic.

2. **Cyber warfare:-** It refers to state-sponsored or nation-level cyber attacks carried out to damage, disrupt, spy on, or control another country's information systems and critical infrastructure.

Against Society

1. **Online gambling** :- It refers to betting or wagering money using internet-based platforms such as websites or mobile apps on games of chance or skill.

2. **Cyber trafficking** :- It refers to the use of the internet and digital platforms to facilitate illegal human trafficking, including recruitment, transportation, exploitation, or sale of individuals.

Common Cybercrime

Phishing: Fake emails/messages tricking users into revealing sensitive info (passwords, card details) or downloading malware.

Identity Theft: Stealing personal data to impersonate someone for financial gain or fraud.

Cyberbullying & Stalking: Using electronic devices to harass, threaten, or follow someone.

Online Fraud: Fake job offers, romance scams, or fake shopping sites to steal money.

Modus operandi (MO) refers to the **method, pattern, or technique used by cybercriminals to plan, execute, and conceal cybercrimes**.

Cybercriminal Modus Operandi (MO)

- **Social Engineering & Deception:** Posing as officials (police, customs) via calls/messages, creating panic, sending fake documents/warrants to coerce victims into paying "deposits" or revealing info.
- **Financial Fraud:** Using aggregator models, creating fake entities to launder money through fintech platforms, demanding payments via UPI, gift cards, or crypto.
- **Identity Theft:** Stealing personal data (passwords via keyloggers in public Wi-Fi, etc.) to impersonate for financial gain or further fraud.
- **Technical Tricks:** Employing advanced malware (APTs), zero-day exploits, creating complex networks (aggregator-on-aggregator) to conceal true identities and evade detection.
- **Money Laundering:** Funneling extorted funds through multiple accounts, often to offshore locations, using digital transactions for quick dispersal.

Reporting of cybercrimes

How to Report Cybercrimes (India-Specific)

- **Immediate Action (Financial Loss):** Call the national helpline 1930 immediately or visit cybercrime.gov.in.
- **File Complaint:** Register on the portal or through the helpline to get an ID, then complete the registration within 24 hours.
- **Contact Local Authorities:** File a First Information Report (FIR) at your nearest police station's Cyber Cell.
- **Bank Involvement:** Inform your bank's customer care for potential fund freezing.
- **Report Suspicious Activity:** Report suspicious websites, social media accounts, or online activities to the portal.

Remedial and mitigation measures

- **Remedial measures** aim to permanently fix a problem's root cause (like patching software or removing contamination), while **mitigation measures** reduce the likelihood or impact of a problem when a full fix isn't possible, acting as a temporary safeguard (like installing a firewall or creating a buffer zone). In essence, mitigation is proactive defense or damage control, whereas remediation is the reactive, complete solution, often used together for comprehensive risk management.

Remedial Measures

Remedial Measures (Fixing the Source)

- **Definition:** Actions that eliminate the underlying issue, making a system or environment fully secure or clean.
- **Examples:**
  - **Cybersecurity:** Applying a software patch, upgrading a system, changing a faulty configuration.
  - **Environmental:** Excavating contaminated soil, completely removing hazardo substances.
  - **Water Damage:** Drying out a property and repairing damaged structures to prevent mold.

Mitigation Measures

Mitigation Measures (Reducing Impact)

- **Definition:** Steps taken to lessen the severity, frequency, or impact of a threat or hazard, often while waiting for a permanent fix.
- Examples:
- **Cybersecurity:** Isolating a vulnerable server, implementing strict access controls, blocking ports.
- **Environmental:** Planting windbreaks to stop pesticide drift, installing vapor barriers over contaminated soil.
- **Water Damage:** Containing water to prevent spread, setting up dehumidifiers to slow damage.

Legal perspective of Cybercrime

Core Legal Frameworks (India Example)

- **Information Technology Act, 2000 (IT Act):** Provides the primary framework for cybercrime, giving legal validity to electronic records and transactions.

 Section 66: Hacking and unauthorized access.

 Section 67: Publishing obscene material electronically.

 Section 69: Government power to intercept data for national security.

 Section 66F: Cyber Terrorism.

- **Indian Penal Code (IPC), 1860:** Addresses traditional crimes adapted to the digital realm, such as:

 Section 420: Cheating and fraud.

 Section 503/506: Criminal intimidation via electronic means.

 Sections 463-471: Forgery.

- **Digital Personal Data Protection Act (DPDP) 2023:** Focuses on data privacy, imposing heavy fines (up to ₹250 crores) for breaches and non-compliance, plus compensation for individuals

Challenges In Enforcement

- **Jurisdictional Issues:** Usually, cybercrimes move among different states, which results in jurisdictional problems. It is almost impossible to investigate and persecute those from other countries.
- **Lack of Awareness:** Many individuals and organizations are not acquainted with cyber laws and their rights. Consequently, there is a small number of cybercrimes that are reported to the authorities, and even preventive measures despite their existence still haven't been the subject of the necessary monitoring processes.
- **Technical Expertise:** The majority of law enforcement agencies do not acquire the technical skills that are vital for them to function properly and, therefore, successfully investigate and prosecute cybercrimes. The issue can be resolved if law enforcement agencies provide the training that is essential to close this particular gap.
- **Evolving Nature of Cybercrime:** The progress in technology moves at such high speed that legislation can hardly catch up with it

Information Technology Act 2000

What is the Information Technology Act 2000?

- A legal framework proposed by the Indian Parliament, the Informational Technology Act of 2000, is the primary legislation in India dealing with cybercrime and electronic commerce.

- It was formulated to ensure the lawful conduct of digital transactions and the reduction of cyber crimes, on the basis of the Model Law on Electronic Commerce adopted by the United Nations Commission on International Trade Law. This legal framework, also known as IT Act 2000, comes with 94 sections, divided into 13 chapters and 4 schedules.

IT Act 2000 and Its Amendments

As technology evolved over time, the Indian Parliament recognised the need to revise the Act in order to align it with societal needs, resulting in its amendment. **Two significant amendments** made to the **IT Act 2000** are as follows:

1. Amendment of 2008

The 2008 amendment came up with modifications to Section 66A of the IT Act, 2000. This section outlined penalties for sharing offensive messages electronically. However, this was omitted by the Jan Vishwas (Amendment of Provisions) Act, 2023. Other significant changes made to the Act by the 2008 amendment was the introduction of electronic signature to authenticate electronic records. It provided that contracts formed through electronic means are valid.

2. Amendment Bill 2015

In 2015, another bill was initiated to amend Section 66A with the aim of safeguarding the fundamental rights guaranteed to citizens by the country's Constitution. This was later accomplished by declaring it as violative of Article 19 of the Constitution and omitting Section 66A from the Act.

Cyber Offenses

A **cyber offence** is any act that violates cyber law.

Common Types of Cyber Offenses

- **Phishing/Smishing/Vishing**: Tricking users into revealing sensitive info via fake emails, texts, or calls.
- **Malware/Ransomware**: Spreading viruses, worms, or ransomware to damage systems or encrypt data for ransom.
- **Identity Theft**: Stealing personal data (passwords, credit cards) to commit fraud.
- **Cyberbullying/Stalking**: Harassment, threats, or repeated unwanted contact via digital means.
- **Hacking**: Unauthorized access to computer systems or networks.
- **Data Breaches**: Stealing or exposing confidential information.
- **Financial Fraud**: Online scams, bank fraud, credit card fraud, e-wallet fraud.
- **Child Exploitation**: Creating or distributing child sexual abuse material (CSAM).
- **Denial of Service (DoS/DDoS) Attacks**: Overwhelming systems to make them unavailable.

Organization dealing with Cybercrimes

In India, the primary organization dealing with cybercrime is the **Indian Cybercrime Coordination Centre (I4C**), a nodal agency under the Ministry of Home Affairs (MHA). The I4C works to coordinate law enforcement efforts, provide a platform for reporting, and build capacity against cyber threats through units like the National Cybercrime Threat Analytics Unit (TAU), National Cybercrime Reporting Portal, and National Cybercrime Training Centre (NCTC). .

Functions of I4C

1. **Coordination:** Acts as the central point for LEAs (Law Enforcement Agencies) to tackle cybercrime.
2. **Reporting:** Manages the National Cyber Crime Reporting Portal (cybercrime.gov.in), where citizens can report crimes, and runs the helpline 1930 for financial fraud.
3. **Analysis:** Uses units like the Threat Analytics Unit (TAU) to identify trends.
4. **Capacity Building:** Develops training and forensic labs through the NCTC and National Cybercrime Forensic Laboratory (NCFL).
5. **Ecosystem Building**: Brings together government, industry, academia, and citizens for a comprehensive fight against cybercrime.

Platforms to deal with Cybercrime

- **Report Online:** Use cybercrime.gov.in to file complaints.
- **Call Helpline:** Dial **1930** for immediate action on financial fraud.
- **Local Police:** Reports are forwarded to the relevant police station for FIR registration and investigation.

Cybersecurity

- **Cyber Security** is the practice of **protecting computers, networks, data, and digital systems** from unauthorized access, attacks, damage, or misuse.

- Cybersecurity is the combination of methods, processes, tools, and behaviors that protect computer systems, networks, and data from **cyberattacks** and unauthorized access.

- Cybersecurity is the practice of defending computers, servers, mobile devices, electronic systems, networks, and data from malicious attacks. It's also known as information technology security or electronic information security.

Importance of Cybersecurity in India

- Protection of Sensitive Data
- Business Continuity and Reputation
- Economic and Regulatory Implications
- National Security and Critical Infrastructure
- Trust and Reputation

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